Winter Weather Panel: Descriptive Report

PRISM county–month data, 1981–2025

2026-08-28

Provenance. All exhibits generated by `codePYTHON/09_descriptive_weather_full.ipynb`, executed on the Kodama server against the complete PRISM county-month panel. Figures are vector PDFs. Tier 1 QA exhibits are indexed in Appendix B rather than included/reproduced here in the report.

Status. The warm-anomaly exhibits in Section 5 are **provisional** pending confirmation of the anomaly definition. Section 8 lists the choices requiring a decision.

Contents

1 Coverage and data integrity

The panel is complete: no duplicate keys, no unmatched rows across the monthly and derived files, no incomplete county-months, and no QA-field mismatches. The excluded county-winters are exactly the two boundary winters that cannot be complete, not unexplained missingness.

Table 1: Coverage and integrity diagnostics

Metric	Value
dataset	PRISM
monthly rows	1678320
derived rows	1678320
merged rows	1678320
counties	3108
first year	1981
last year	2025
duplicate monthly keys	0
duplicate derived keys	0
monthly only rows	0
derived only rows	0
incomplete months	0
qa mismatch n days	0
qa mismatch expected days	0
qa mismatch is incomplete	0
qa mismatch dataset types	0
candidate county winters	142968
complete county winters	136752
excluded incomplete county winters	6216
first complete winter	1982
last complete winter	2025
expected calendar years	1981–2025
observed calendar years	1981–2025
observed complete winters	1982–2025

Table 2: Tier 1 QA findings

Check	Result	Status
Duplicate county-year-month keys	0	PASS
Rows appearing in only one merge source	0	PASS
Incomplete county-months	0	PASS
Excluded incomplete county-winters	6216	INFO
County-variable mean flags ($\|z\| \geq 5$)	20	REVIEW

2 Long-run winter warming

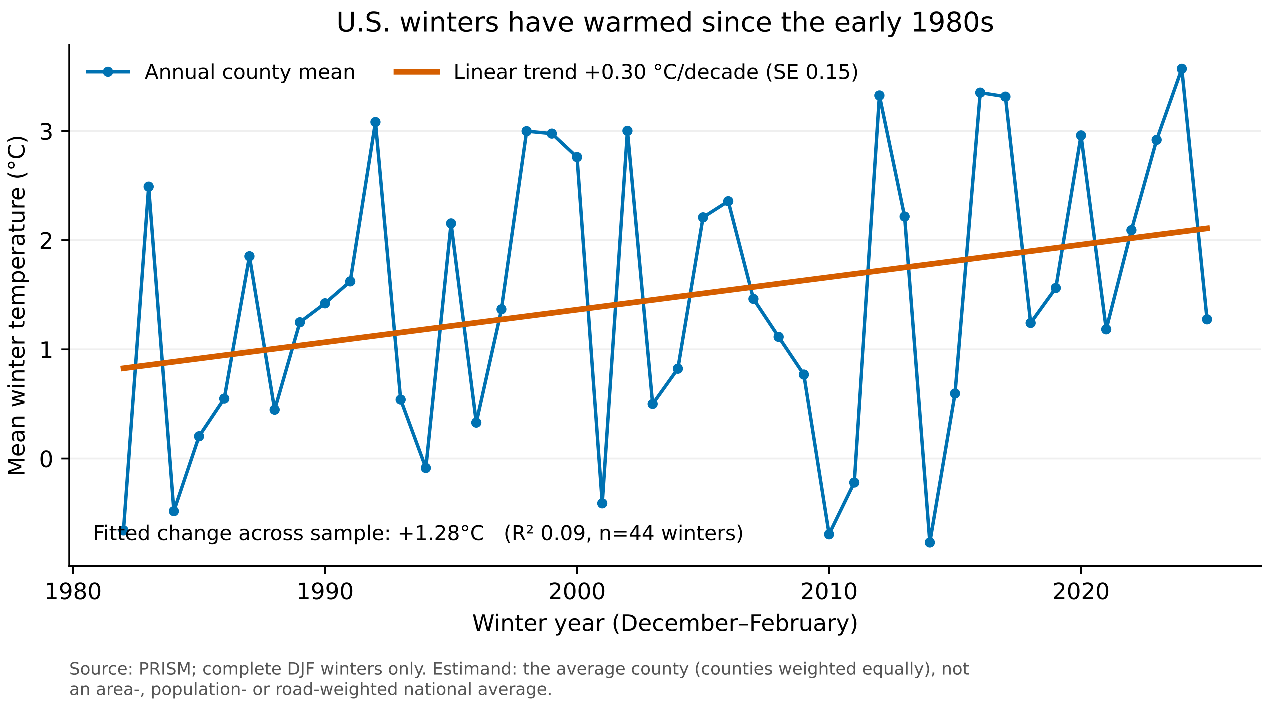


Figure 1: National winter temperature, 1982–2025. Each point is the unweighted mean across counties, so the estimand is the average county rather than an area- or population-weighted national average. The fitted slope is shown with its standard error.

State-level trends are reproduced in Appendix A.

3 Where winters have warmed

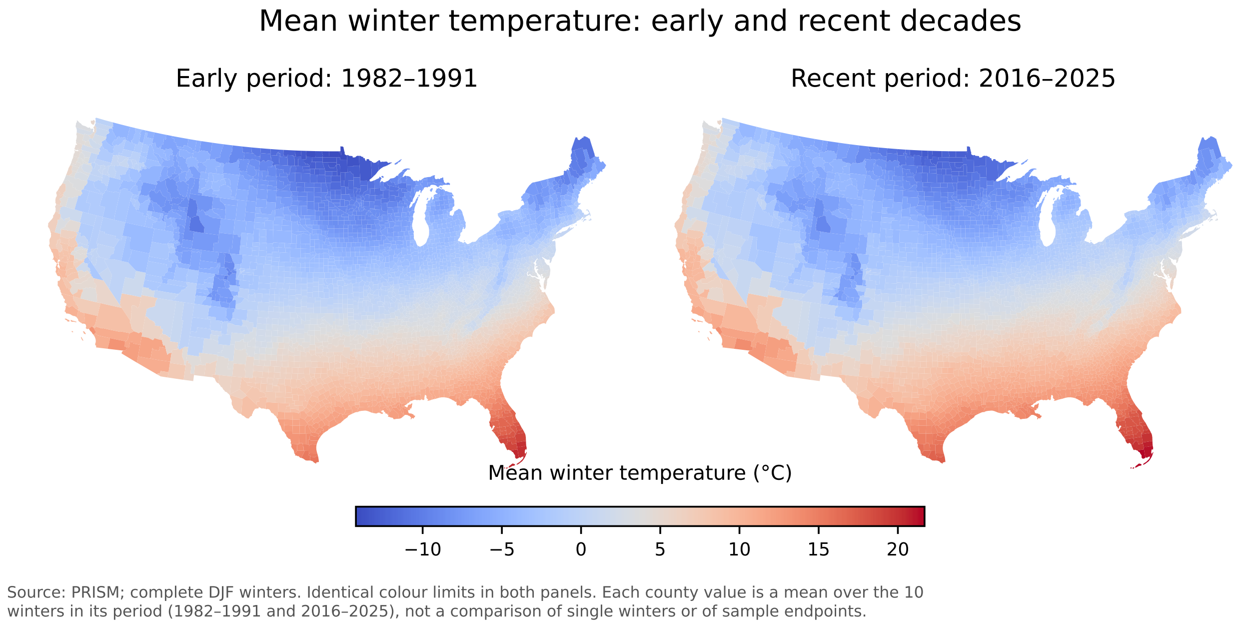


Figure 2: Mean winter temperature in the early (1982–1991) and recent (2016–2025) periods, on identical colour limits. Both panels are ten-winter period means, not single-year endpoints.

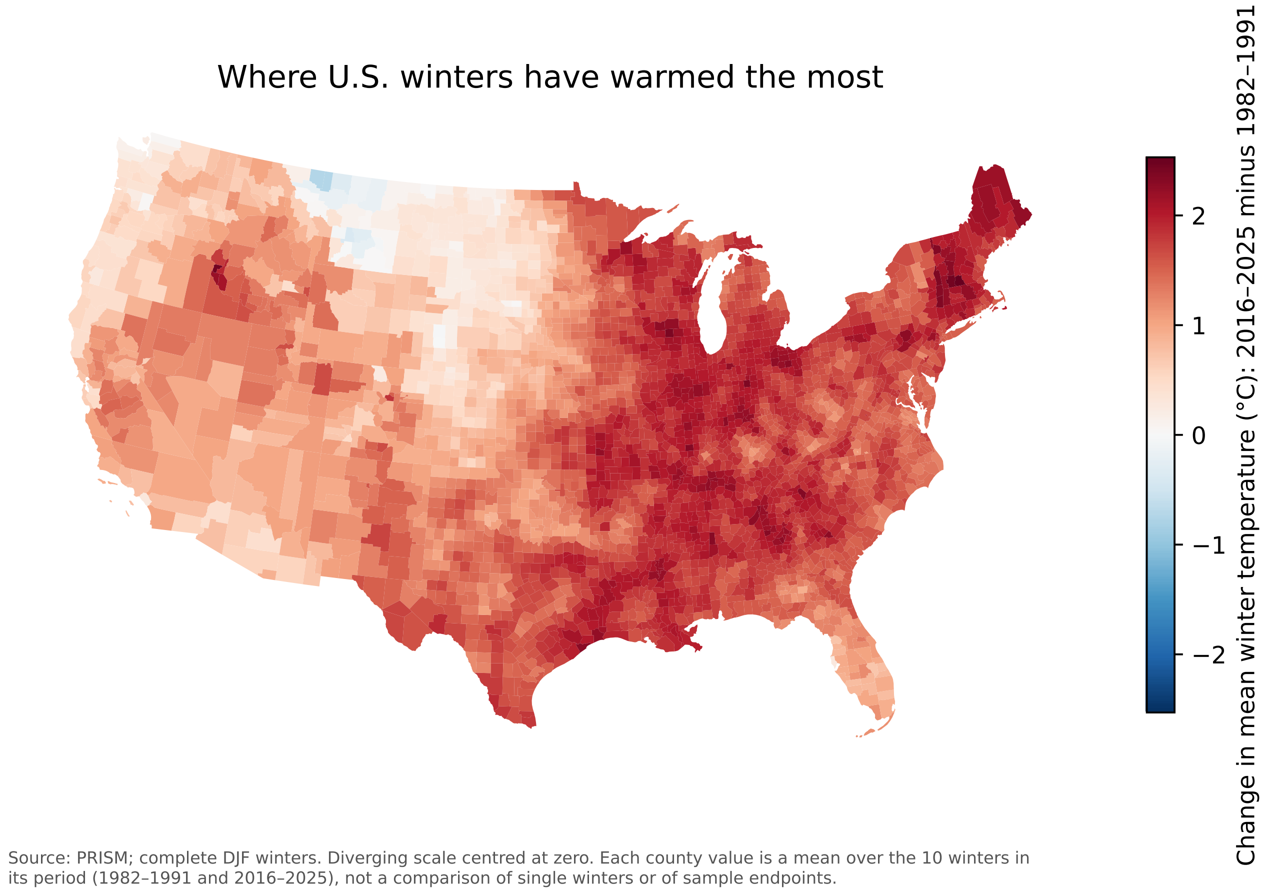


Figure 3: Change in mean winter temperature between the two periods. The diverging scale is centred at zero, so the map shows both the breadth of warming and its regional heterogeneity.

4 Cold extremes

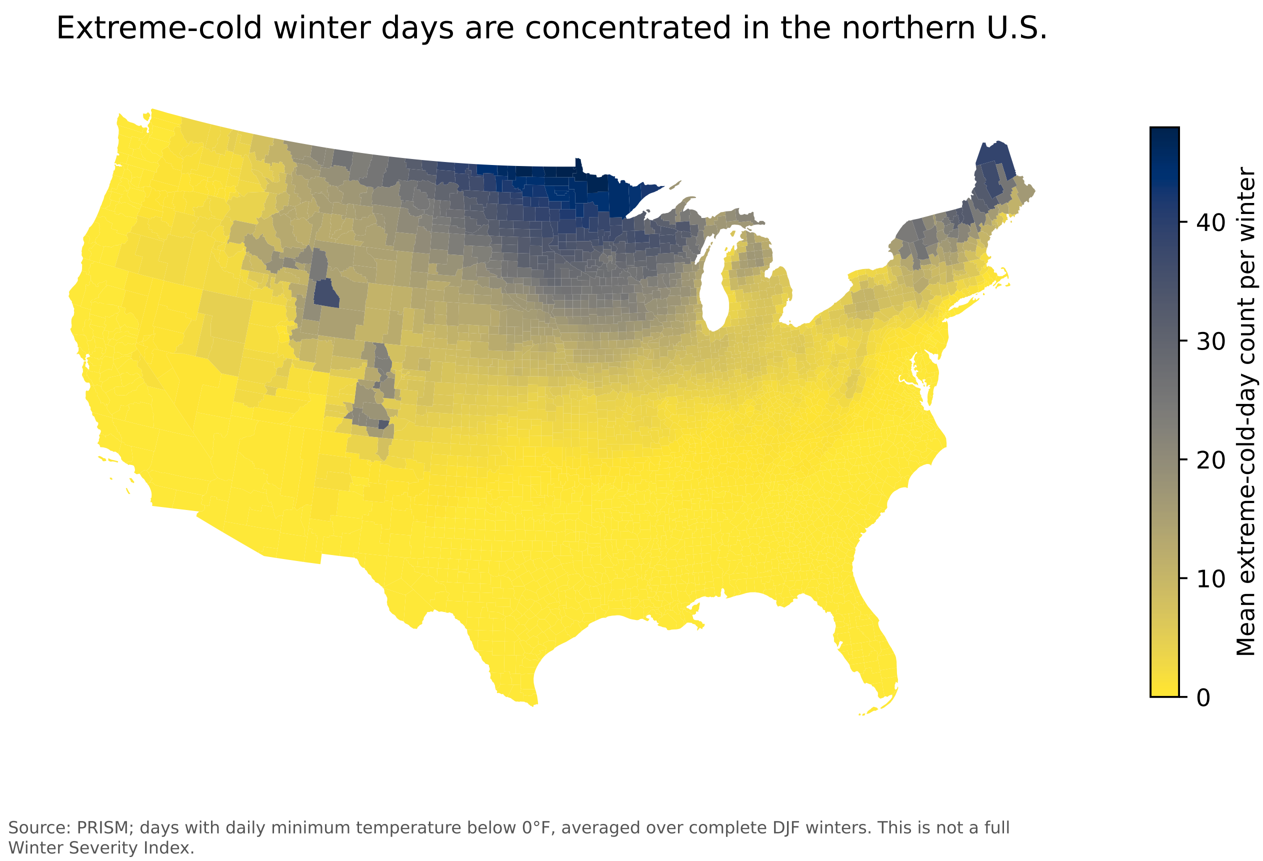
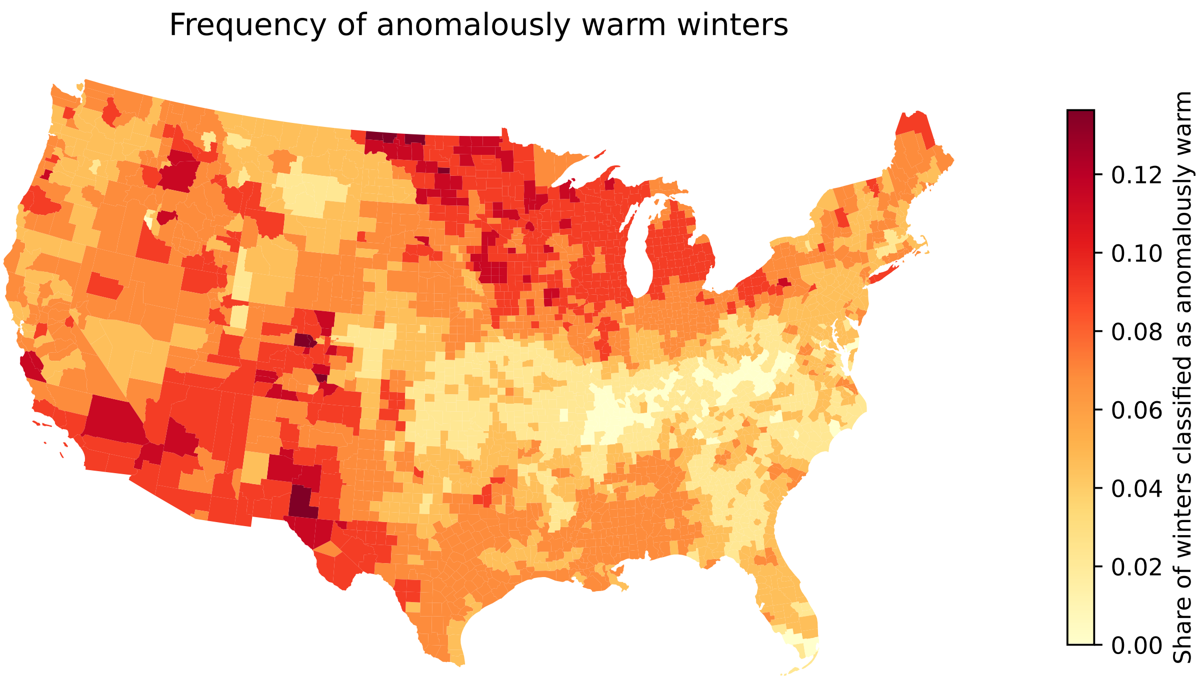


Figure 4: Mean count of winter days with a daily minimum below 0°F. This is the extreme-cold measure only; it is not a Winter Severity Index.

5 Anomalously warm winters



PROVISIONAL — requires Eyal's confirmation. County-specific detrended residual > 1.5 SD; at least 30 complete winters. The definition uses the full sample to classify each winter and removes each county's trend, so mapped shares partly reflect residual tail shape. See the caveat table.

Figure 5: Provisional. Share of a county's eligible winters classified as anomalously warm, defined as a detrended residual above 1.5 standard deviations of that county's own full-sample distribution.

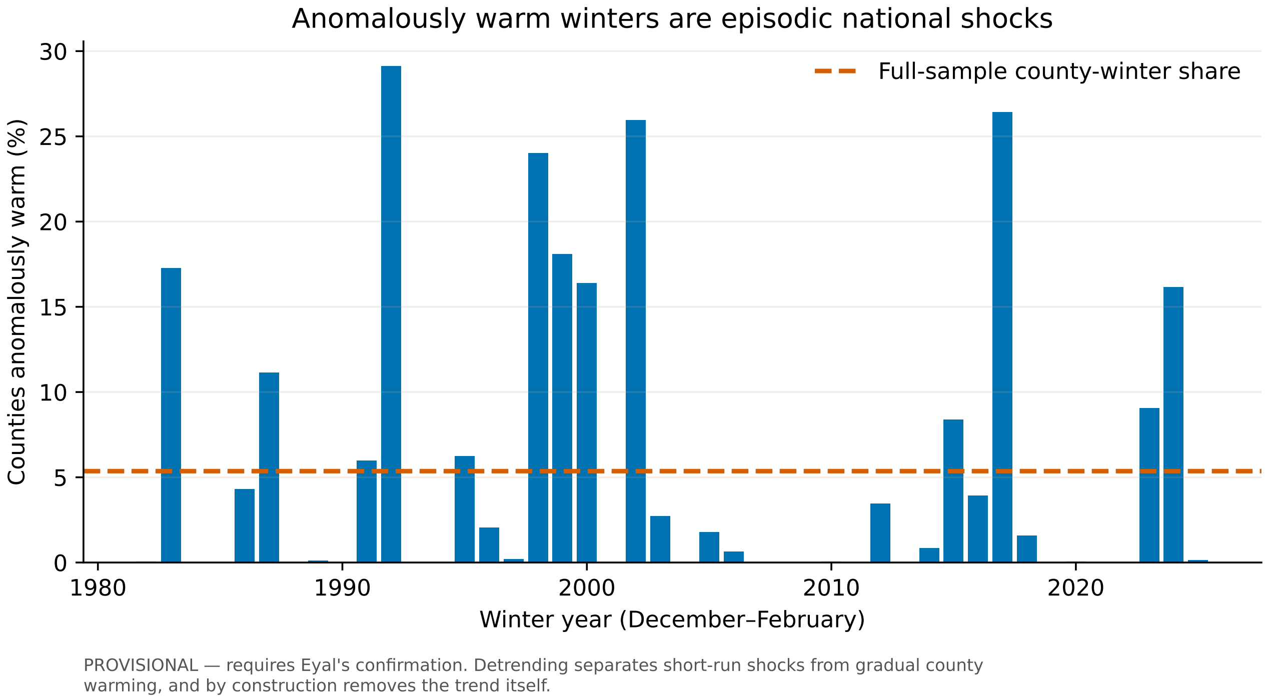


Figure 6: Provisional. Share of counties classified as anomalously warm in each winter. Because each county's linear trend is removed, this isolates episodic shocks from gradual warming.

Known properties of the provisional definition

1. Look-ahead: each county's reference distribution uses the full 1982–2025 sample, so later winters inform how earlier winters are classified.

2. Detrending removes each county's linear trend, so gradual warming is excluded from the shock measure by construction.

3. A county's mapped frequency partly reflects the skewness and tail shape of its residual distribution, not exposure to warm shocks alone.

4. The 1.5 SD cut-off is a convention, not an estimated threshold.

5. For regression use, compare against a fixed-baseline alternative (e.g. above the county's 1982–2000 mean by 1.5 baseline SD) or use the continuous standardised anomaly.

6 Distributional shift

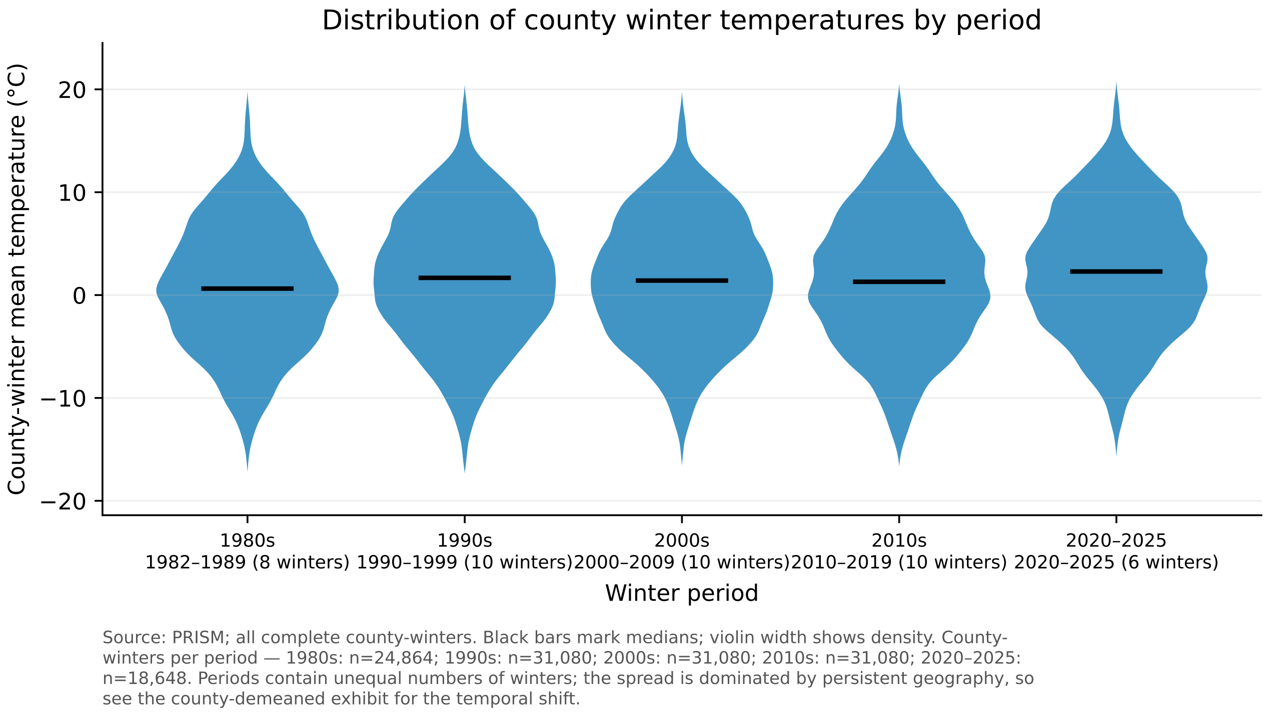
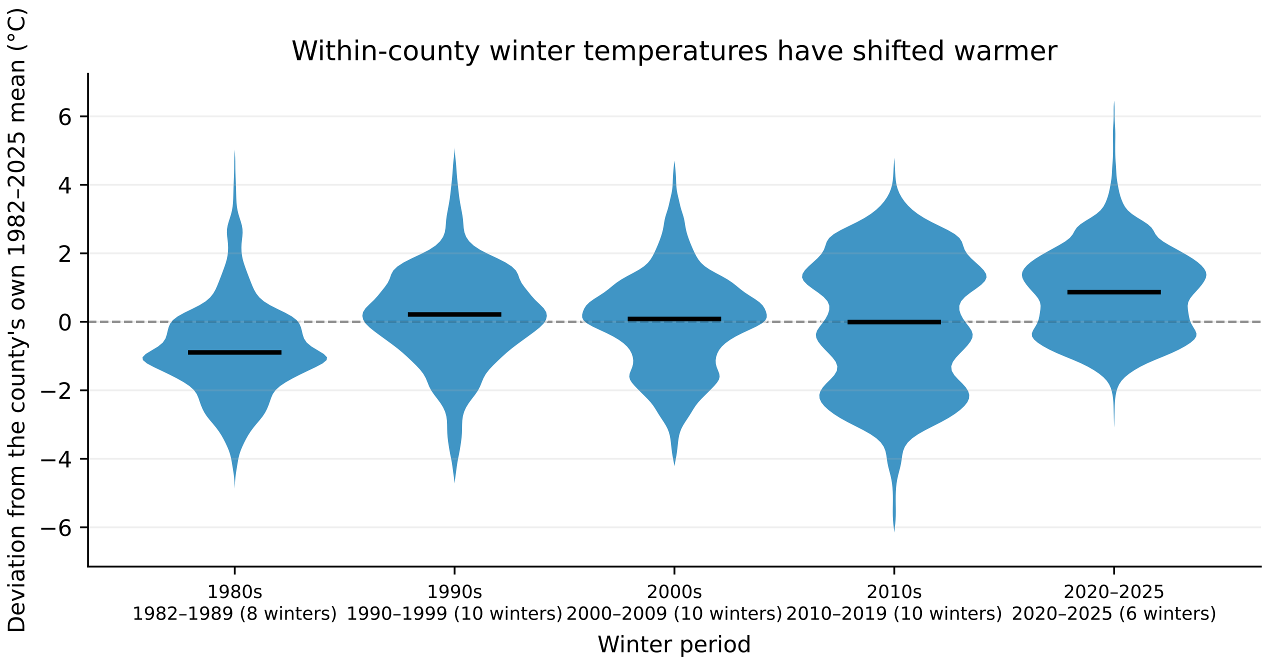


Figure 7: County-winter mean temperatures pooled by period. Periods contain unequal numbers of winters, and the pooled spread is dominated by persistent cross-county geography, which compresses the apparent temporal shift.



Source: PRISM; all complete county-winters, each expressed as a deviation from its own county's full-sample mean. Removing persistent geography isolates the temporal shift. County-winters per period — 1980s: n=24,864; 1990s: n=31,080; 2000s: n=31,080; 2010s: n=31,080; 2020–2025: n=18,648.

Figure 8: The same periods after subtracting each county's own full-sample mean. Removing persistent geography isolates the within-county shift, which is the quantity of interest.

7 Robustness: PRISM versus ERA5

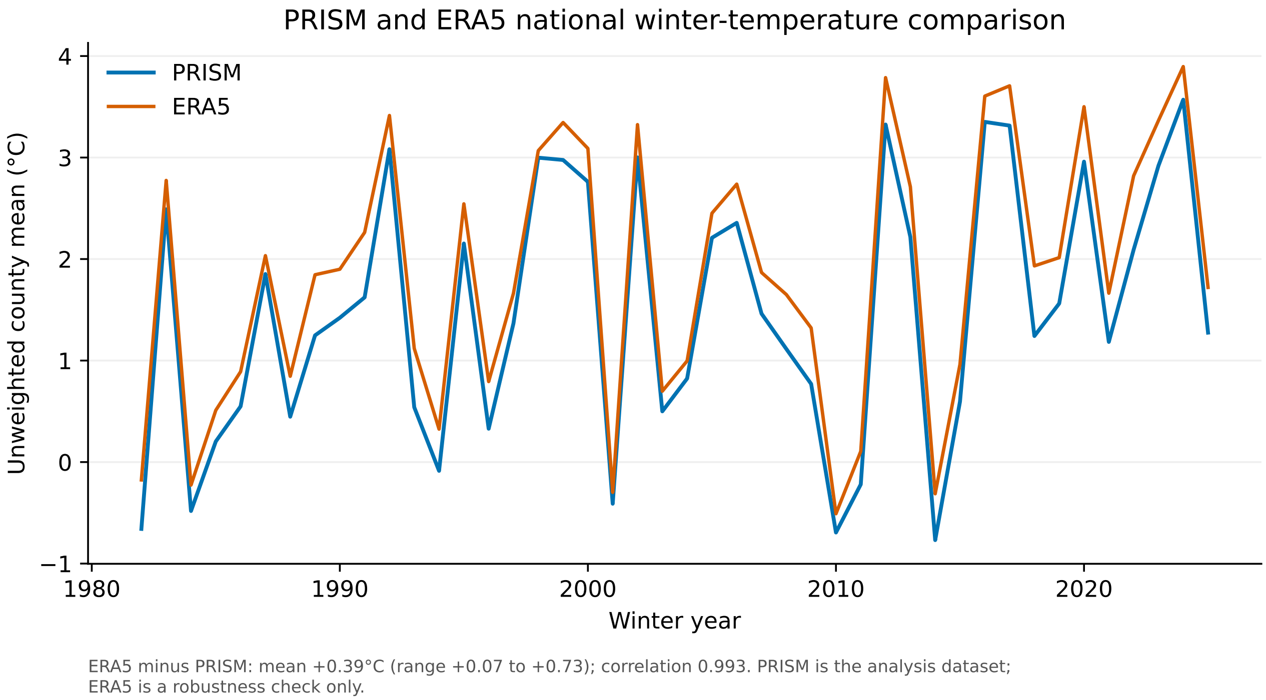


Figure 9: National winter temperature from both products. PRISM is the analysis dataset; ERA5 is a targeted robustness check. The mean offset and correlation are reported on the figure.

8 Decisions requested

Anomaly definition (regression use)

Current choice. Detrended county residual > 1.5 SD of the full 1982–2025 sample; minimum 30 winters

Issue. Uses future winters to classify earlier ones and removes secular warming by construction; threshold is a convention. Alternatives: fixed 1982–2000 baseline, or the continuous standardised anomaly.

Status. CONFIRM — descriptive exhibits are labelled PROVISIONAL

Geographic weighting

Current choice. Unweighted mean across counties

Issue. Estimand is the average county, not an area-, population- or road-weighted national average. A VMT- or road-mile-weighted series may be the relevant exposure weight for the collisions stage.

Status. CONFIRM

Comparison periods

Current choice. 1982–1991 versus 2016–2025

Issue. Symmetric ten-winter period means; not 1981-versus-2025 endpoints.

Status. CONFIRM

Extreme-cold exhibit

Current choice. Mean days with TMIN < 0°F; not labelled WSI

Issue. Not a full Winter Severity Index.

Status. DOCUMENTED

ERA5 robustness exhibit

Current choice. Targeted national PRISM-ERA5 comparison

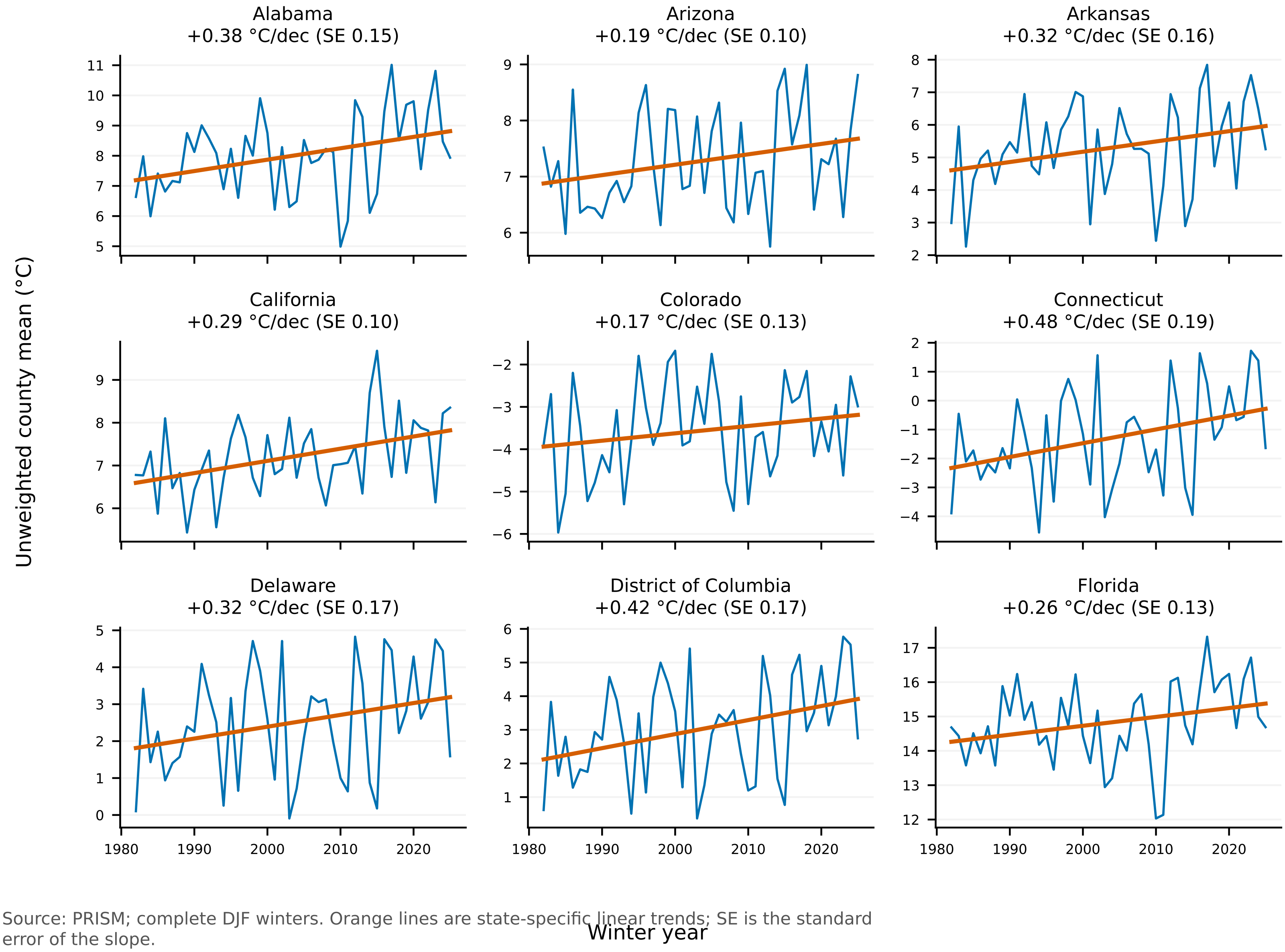
Issue. PRISM remains the analysis dataset; ERA5 runs warm by a small constant offset.

Status. DOCUMENTED

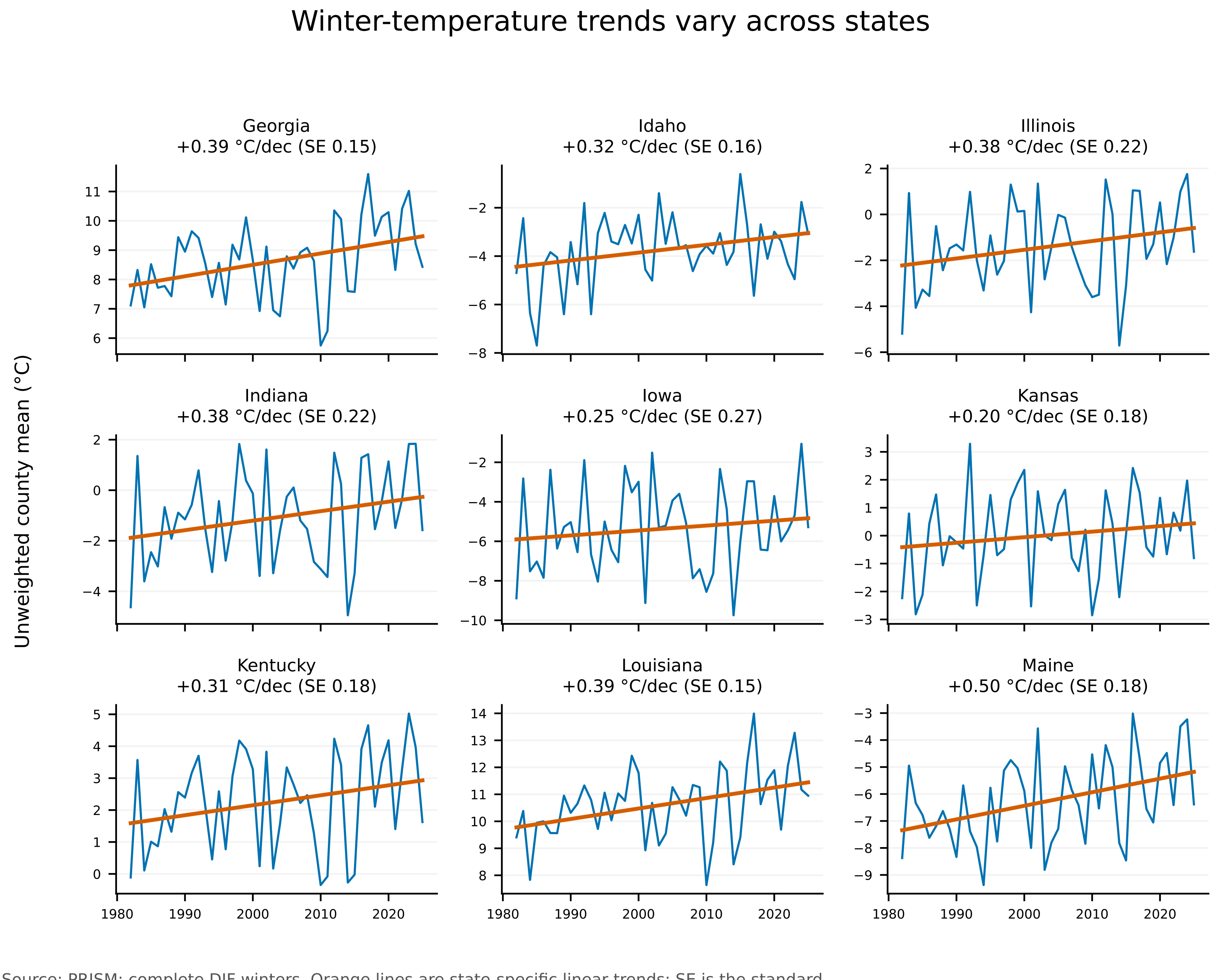
A State-level winter-temperature trends

Per-state annual means with state-specific linear trends and the standard error of each slope. Reproduced in full; page count follows the run.

Winter-temperature trends vary across states



Source: PRISM; complete DJF winters. Orange lines are state-specific linear trends; SE is the standard error of the slope.



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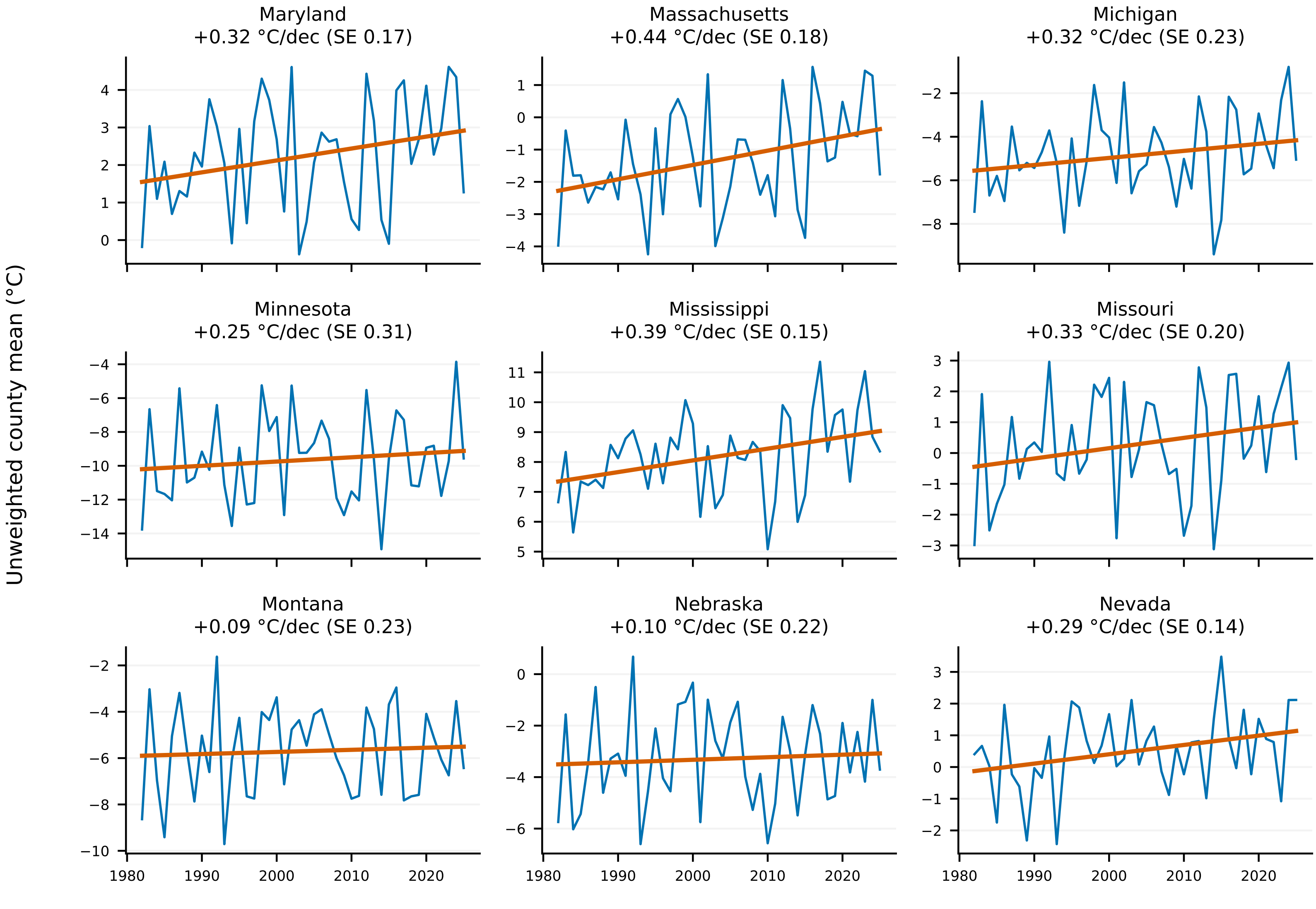

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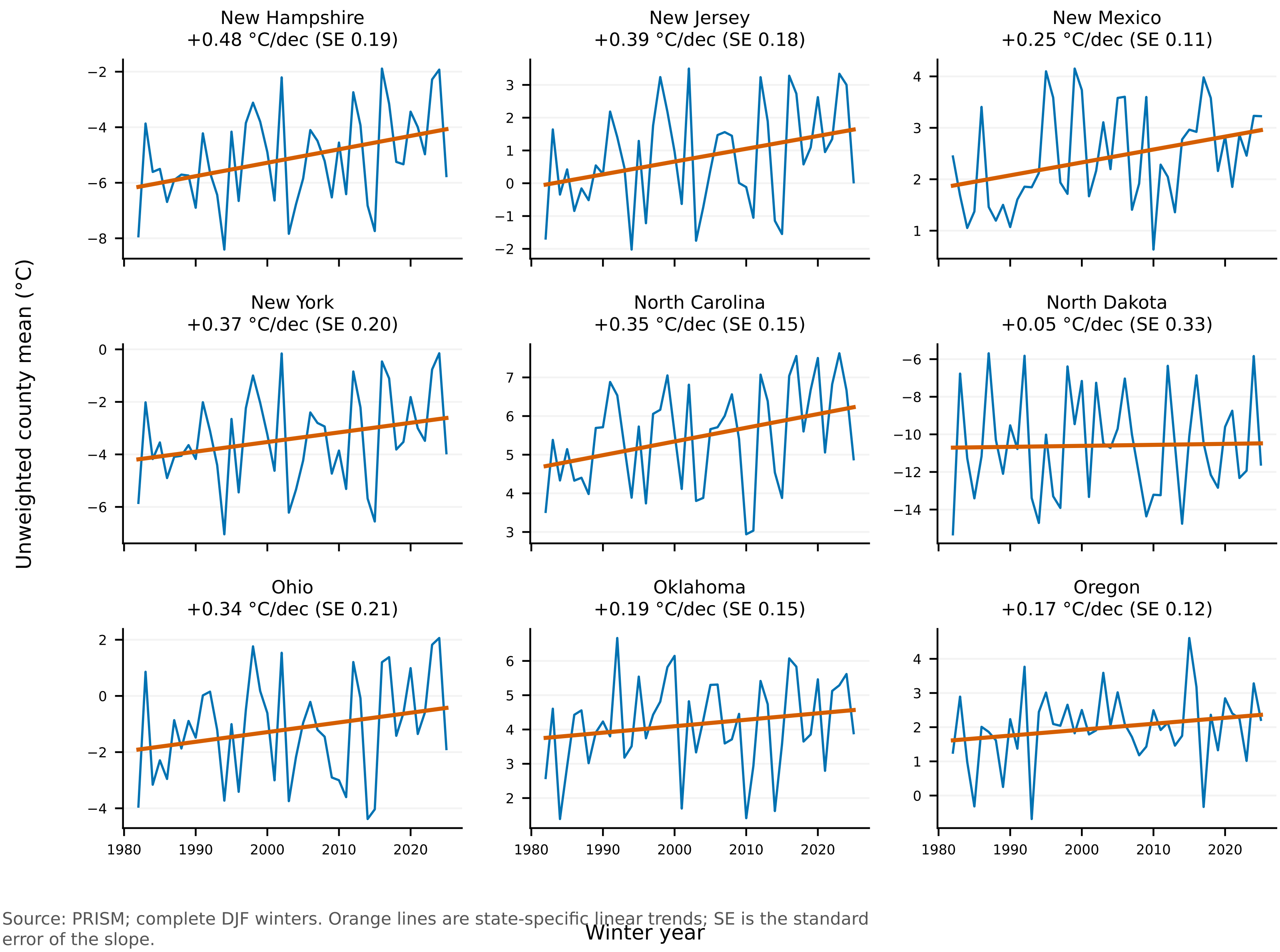

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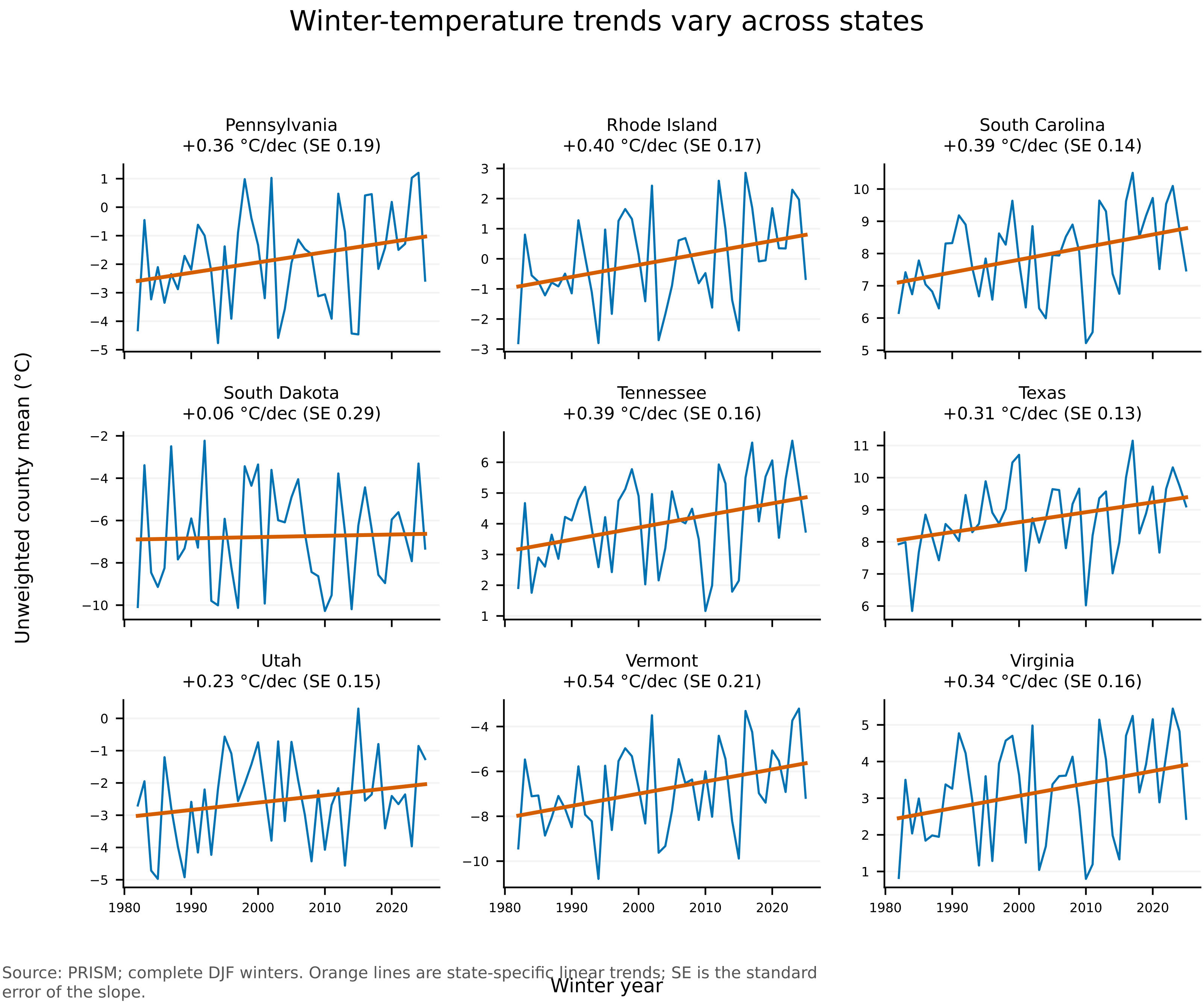

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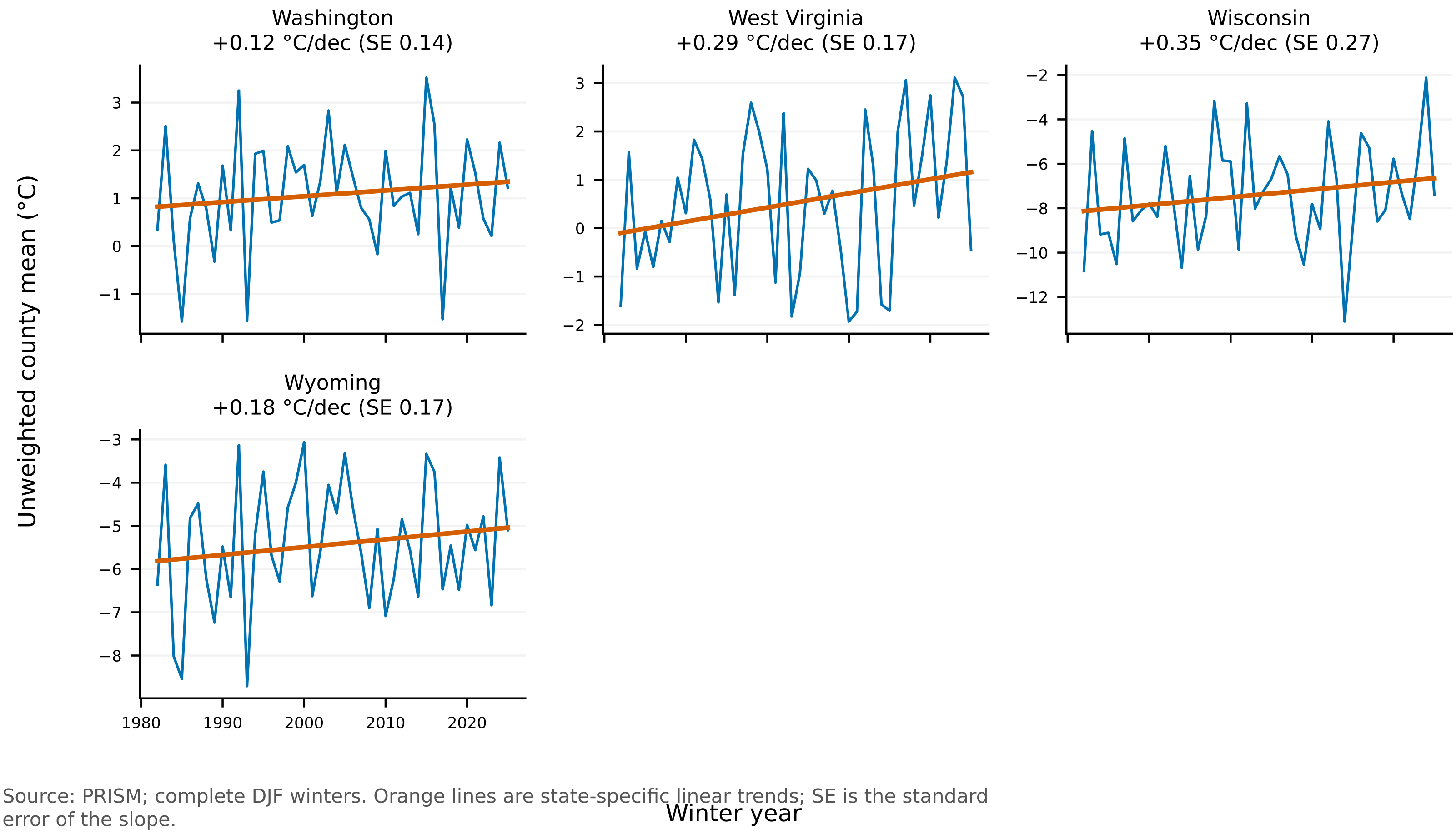

Source: PRISM; complete DJF winters. Orange lines are state-specific linear trends; SE is the standard error of the slope.

B Tier 1 QA artefacts (not reproduced)

The following internal QA exhibits and tables were produced by the same run but not incuded in this report.

- `figures/weather/tier1/prism_interannual_variability_choropleth.pdf`
- `figures/weather/tier1/prism_national_trend_days_below_freezing_32f.pdf`
- `figures/weather/tier1/prism_national_trend_days_extremely_cold.pdf`
- `figures/weather/tier1/prism_national_trend_freeze_thaw_days.pdf`
- `figures/weather/tier1/prism_county_spaghetti_mean_temp_c.pdf`
- `figures/weather/tier1/prism_state_trends_freeze_thaw_days.pdf`
- `figures/weather/tier1/prism_state_decade_distributions_mean_temp_c.pdf`
- `figures/weather/tier1/prism_state_trends_mean_temp_c.pdf`
- `figures/weather/tier1/prism_state_trends_days_below_freezing_32f.pdf`
- `figures/weather/tier1/prism_national_trend_mean_temp_c.pdf`
- `figures/weather/tier1/prism_state_trends_days_extremely_cold.pdf`
- `tables/weather/tier1/prism_tier1_qa_findings.csv`
- `tables/weather/tier1/prism_summary_by_county_review_flags.csv`
- `tables/weather/tier1/prism_summary_pooled.csv`
- `tables/weather/tier1/prism_interannual_variability_by_state.csv`
- `tables/weather/tier1/prism_summary_by_calendar_month.csv`
- `tables/weather/tier1/prism_tier1_coverage_integrity.csv`
- `tables/weather/tier1/prism_summary_by_county.csv`
- `tables/weather/tier1/prism_county_interannual_variability.csv`